

Product Review and Rating Management System

PURPLE COMPANY E-COMMERCE

1. Create Review and Rating Tables

1.1 PurpleReview

Column	Type	Constraint
Review_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Review_Text	VARCHAR2(500)	—
Review_Date	DATE	—

1.2 PurpleRating

Column	Type	Constraint
Rating_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Rating	NUMBER	—
Rating_Date	DATE	—

2. PurpleReview Table

```
SQL> CREATE TABLE PurpleReview (  
2     REVIEW_ID NUMBER PRIMARY KEY,  
3     CUSTOMER_ID NUMBER,  
4     PRODUCT_ID NUMBER,  
5     REVIEW_TEXT VARCHAR2(500),  
6     REVIEW_DATE DATE  
7 );
```

Table created.

3. Rating Table

```
SQL> CREATE TABLE PurpleRating (  
2     RATING_ID NUMBER PRIMARY KEY,  
3     REVIEW_ID NUMBER,  
4     RATING NUMBER(1)  
5 );  
  
Table created.
```

4. Review Data Insertion

```
SQL> INSERT INTO PurpleReview  
2 (REVIEW_ID, CUSTOMER_ID, PRODUCT_ID, REVIEW_TEXT, REVIEW_DATE)  
3 VALUES  
4 (1, 1, 101, 'Good product and nice quality', SYSDATE);  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO PurpleReview  
2 (REVIEW_ID, CUSTOMER_ID, PRODUCT_ID, REVIEW_TEXT, REVIEW_DATE)  
3 VALUES  
4 (2, 2, 102, 'Very useful and worth the price', SYSDATE);  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO PurpleReview  
2 (REVIEW_ID, CUSTOMER_ID, PRODUCT_ID, REVIEW_TEXT, REVIEW_DATE)  
3 VALUES  
4 (3, 3, 103, 'Quality is excellent', SYSDATE);  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO PurpleReview  
2 (REVIEW_ID, CUSTOMER_ID, PRODUCT_ID, REVIEW_TEXT, REVIEW_DATE)  
3 VALUES  
4 (4, 4, 104, 'Average product', SYSDATE);  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO PurpleReview  
2 (REVIEW_ID, CUSTOMER_ID, PRODUCT_ID, REVIEW_TEXT, REVIEW_DATE)  
3 VALUES  
4 (5, 5, 105, 'Really satisfied with the product', SYSDATE);  
  
1 row created.
```

4.1 Display Review Records

```
SQL> SELECT * FROM PurpleReview;
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
1 1 101
```

```
Good product and nice quality
```

```
07-OCT-26
```

```
2 2 102
```

```
Very useful and worth the price
```

```
07-OCT-26
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
3 3 103
```

```
Quality is excellent
```

```
07-OCT-26
```

```
4 4 104
```

```
Average product
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
07-OCT-26
```

```
5 5 105
```

```
Really satisfied with the product
```

```
07-OCT-26
```

5. Rating Data Insertion

```

SQL> INSERT INTO PurpleRating VALUES (1, 1, 5);
1 row created.

SQL> INSERT INTO PurpleRating VALUES (2, 2, 4);
1 row created.

SQL> INSERT INTO PurpleRating VALUES (3, 3, 5);
1 row created.

SQL> INSERT INTO PurpleRating VALUES (4, 4, 4);
1 row created.

SQL> INSERT INTO PurpleRating VALUES (5, 5, 5);
1 row created.

SQL>
SQL> COMMIT;

Commit complete.

```

5.1 Display Rating Records

```

SQL> SELECT * FROM PurpleRating;

```

RATING_ID	REVIEW_ID	RATING
-----	-----	-----
1	1	5
2	2	4
3	3	5
4	4	4
5	5	5

6.Retrieve Product Review Details

```

SQL> SELECT
  2     p.Product_Name,
  3     c.Customer_Name,
  4     r.Review_Text,
  5     rt.Rating,
  6     r.Review_Date
  7 FROM PurpleReview r
  8 JOIN PurpleCustomer c
  9     ON r.Customer_ID = c.Customer_ID
 10 JOIN PurpleProduct p
 11     ON r.Product_ID = p.Product_ID
 12 JOIN PurpleRating rt
 13     ON r.Review_ID = rt.Review_ID;

```

no rows selected

```
SQL> SELECT * FROM PurpleReview;
```

```

  REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----

```

```

  REVIEW_TEXT
-----

```

```

  REVIEW_DA
-----

```

```

          1          1          101
Good product and nice quality
07-OCT-26

```

```

          2          2          102
Very useful and worth the price
07-OCT-26

```

```

  REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----

```

```

  REVIEW_TEXT
-----

```

```

  REVIEW_DA
-----

```

```

          3          3          103

```

```

REVIEW_TEXT
-----
REVIEW_DA
-----
07-OCT-26

          5          5          105
Really satisfied with the product
07-OCT-26

SQL> SP2-0042: unknown command "S" - rest of line ignored.
SQL> SELECT * FROM PurpleProduct;

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND                                     PRICE      STOCK
-----
CATEGORY_ID
-----
          1
Matte Lipstick
Purple                                     499         50
          1

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND                                     PRICE      STOCK
-----
CATEGORY_ID
-----
          2
Liquid Foundation
Purple                                     799         40
          2

```

7. Calculate Average Product Rating

```

SQL> SELECT
2      p.Product_Name,
3      AVG(rt.Rating) AS Average_Rating
4  FROM PurpleProduct p
5  JOIN PurpleReview r
6      ON p.Product_ID = r.Product_ID
7  JOIN PurpleRating rt
8      ON r.Review_ID = rt.Review_ID
9  GROUP BY p.Product_Name;

```

no rows selected

```
SQL> SELECT * FROM PurpleRating;
```

RATING_ID	REVIEW_ID	RATING
1	1	5
2	2	4
3	3	5
4	4	4
5	5	5

8. Identify Highly Rated Products

```

SQL> SELECT
2      p.Product_ID,
3      p.Product_Name,
4      rt.Rating
5  FROM PurpleProduct p
6  JOIN PurpleReview r
7      ON p.Product_ID = r.Product_ID
8  JOIN PurpleRating rt
9      ON r.Review_ID = rt.Review_ID
10 WHERE rt.Rating >= 4;

```

no rows selected

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME
2  FROM PurpleProduct;
```

PRODUCT_ID

PRODUCT_NAME

1
Matte Lipstick

2
Liquid Foundation

3
Waterproof Eyeliner

PRODUCT_ID

PRODUCT_NAME

4
Glow Blush

5
Volume Mascara

```
SQL> UPDATE PurpleReview SET Product_ID = 1 WHERE Review_ID = 1;  
1 row updated.
```

```
SQL> UPDATE PurpleReview SET Product_ID = 2 WHERE Review_ID = 2;  
1 row updated.
```

```
SQL> UPDATE PurpleReview SET Product_ID = 3 WHERE Review_ID = 3;  
1 row updated.
```

```
SQL> UPDATE PurpleReview SET Product_ID = 4 WHERE Review_ID = 4;  
1 row updated.
```

```
SQL> UPDATE PurpleReview SET Product_ID = 5 WHERE Review_ID = 5;  
1 row updated.
```

```
SQL>
```

```
SQL> COMMIT;
```

```
Commit complete.
```

```
SQL> SELECT * FROM PurpleRating;
```

RATING_ID	REVIEW_ID	RATING
1	1	5
2	2	4
3	3	5
4	4	4
5	5	5